goals even when this harms the greater good. Bureaucrats tasked with accomplishing a narrow mission may be ignorant of the wider impact of their actions, and it has always been tricky to ensure that their actions remain aligned with the greater good of society. When armies operate along bureaucratic lines – as all modern armies do – it creates a huge gap between a captain commanding a company in the field and the president formulating long-term policy in a distant office. The captain is prone to make decisions that seem reasonable on the ground but that actually undermine the war's ultimate goal.

We see, then, that the alignment problem has long predated the computer revolution and that the difficulties encountered by builders of present-day information empires are not unlike those that bedevilled previous would-be conquerors. Nevertheless, computers do change the nature of the alignment problem in important ways. No matter how difficult it used to be to ensure that human bureaucrats and soldiers remain aligned with society's long-term goals, it is going to be even harder to ensure the alignment of algorithmic bureaucrats and autonomous weapon systems.

THE PAPER-CLIP NAPOLEON

One reason why the alignment problem is particularly dangerous in the context of the computer network is that this network is likely to become far more powerful than any previous human bureaucracy. A misalignment in the goals of superintelligent computers might result in a catastrophe of unprecedented magnitude. In his 2014 book *Superintelligence*, the philosopher Nick Bostrom illustrated the danger using a thought experiment, which is reminiscent of Goethe's 'Sorcerer's Apprentice'. Bostrom asks us to imagine that a paper-clip factory buys a superintelligent computer and that the factory's human manager gives the computer a seemingly simple task: produce as many paper clips as possible. In pursuit of this goal, the paper-clip computer conquers the whole of planet Earth, kills all the humans, sends expeditions to take over additional planets and uses the enormous resources it acquires to fill the entire galaxy with paper-clip factories.

The point of the thought experiment is that the computer did exactly what it was told (just like the enchanted broomstick in Goethe's poem). Realising